Project Title: Portable Hemodialysis Machine

BME 3200: Biomedical Systems Design Project

A project submitted *to the Department of Biomedical Engineering in Third Year Second* Semester

Supervised by	**Submitted by**
Name: Jeba Fairooz Rahman	Name: Husain Hamid Chowdhury
Lecturer	1915026
Department of Biomedical Engineering	Department of Biomedical Engineering

Department of Biomedical Engineering

Khulna University of Engineering & Technology

Khulna-9203, Bangladesh

November,2023

Abstract

Dedicated to crafting an innovative, cost-effective portable hemodialysis system, this project aims to revolutionize care for individuals dealing with kidney disorders. The primary goal is to engineer a user-friendly solution prioritizing enhanced patient mobility, treatment efficacy, and affordability. Meticulous engineering and technological advancements ensure both portability and adaptability across diverse healthcare settings, maintaining efficiency while minimizing costs.

Central to this solution is its focus on achieving cost-effectiveness without compromising care quality. Employing innovative design strategies and leveraging technological advancements, the system aims to significantly reduce the economic burden on patients and healthcare institutions. Its compact form factor and enhanced mobility redefine dialysis care standards, offering an accessible, efficient, and economically viable alternative.

The project details the iterative design process, technological innovations, and rigorous validation procedures undertaken to ensure the system's efficacy and compliance with healthcare standards. Envisioned to markedly improve patient quality of life while providing a cost-effective and reliable solution, this portable hemodialysis system represents a transformative step forward in dialysis care, promising significant advancements in patient treatment accessibility and affordability.

List of Abbreviations & Symbols

wt %	Weight Percent
°C	Degree Celsius
MJ/kg	Mega joule per Kilogram
Kg/m^3	Kilogram per cubic meter
KgOE	Kilogram of Oil Equivalent
GCV	Gross Calorific value
SBR	Styrene Butadiene Rubber
BR	Butadiene Rubber
IR	Isoprene Rubber
HDPE	High Density Poly Ethylene
LDPE	Low Density Poly Ethylene
PP	Poly Propylene
PVC	Poly Vinyl Choloride
PET	Poly Ethelene Terapthalate
PS	Poly Styrene
EPT	Ethylene Propylene Terpolymer
IIR	Isobutylene Iso-prene Rubber
UV	Ultraviolet
NR	Natural Rubber
PBR	Polybutadiene Rubber
EPA	European Pollution Authority
MJ	Mega Joule
FCC	Fluid Catalytic Craking
TGA	Thermo Gravimetric Analysis
KW	Kilowatt
N2	Nitrogen Gas
CT ratio	Catalyst -Tyre ratio
ASTM	American Society for Testing Materials
FTIR	Fourier Transform Infra –Red
CARS	Centre for Advanced Research in Sciences
TCR	Total Capital Requirement
FCI	Fixed Capital Investment.
O-H	Hydroxyl stretching
N-H	Amines
C-H	Carbon Hydrogen (Alkanes group)
C=C	Carbon Hydrogen double bonding
C-O	Carbonyl Stretching
O-H	Alcohol, Phenol group
C≡C	Carbon Hydrogen Triple Bonding (Alkynes)

Index

CHAPTER 1	**Introduction**	10
	1.1 Background of project	10
	1.2 Objectives of this project	10
CHAPTER 2	**Motivation/ Background Study**	07
	2.1 Urinary system	08
	2.2 Mechanism of Wastages (Urine) Formation	09
	2.2.1 Glomerular filtration	09
	2.2.2 Reabsorption	10
	2.2.3 Secretion	11
	2.3 Dialysis	11
	2.3.1 History of dialysis	11
	2.4 Types of dialysis	12
	2.4.1 Hemodialysis	12
	2.4.2 Peritonial dialysis	13
CHAPTER 3	**Methodology**	14
	3.1 Apparatus Required	14
	3.2 Architecture of the hemodialysis machine	15
	3.3 Flowchart	16
CHAPTER 4	**Design procedure/Experimental set-up/ Circuit Diagram**	17
	4.1 Design Procedure	17
	4.2 Circuit Design	17
CHAPTER 5	**Implementation**	19
	5.1 Circuit Implementation (Step by Step)	19
	5.2 Circuit Implementation for the Main System	19
	5.3 Implementation of the extended part	22
CHAPTER 6	**Results and Discussions**	23
	6.1 Results	23
	6.1.1 Main Circuit	23
	6.1.2 Extended Circuit	23
	6.2 Discussion	24
CHAPTER 7	**Conclusion and Future Work**	25
	7.1 Conclusion	25
	7.2 Future Work	25
	References	26
	Appendix A	27
	Appendix B	31

LIST OF TABLES

Table No.	Description	Page
3.1	Components required for the System	14

LIST OF FIGURES

Table No.	Description	Page
1.1	Urinary System	9
2.2	Filtration in Nephron	10
2.3	Tubular Reabsorption in the Nephron.	11
2.4	Tubular Secretion in Nephron	12
2.5	Dr. Georg Haas performing dialysis on a patient at the University of Giessen	13
2.6	Hemodialysis Process	14
2.7	Peritonial Dialysis Process	14
3.1	Functional block diagram of the Hemodialysis Machine.	16
3.2	Desired Flowchart of the System	17
4.1	Design Procedure of the System	18
4.2	Overall circuit diagram of the system,(a) for the dialysate mixing and pumping process (b) fir measuring the heart rate and the arterial, venous pressure.	18

5.1	Circuit for the dialysate mixing and pumping process	19
5.2	Implementation of the smart Heparin pump using servo motor.	20
5.3	Setup of the Dialyzer to the system.	21
5.4	Final implementation of the project (front view).	21
5.5	Final implementation of the system (back view).	22
5.6	Extended part that measure the heart rate and vascular pressure.	22
6.1	Output of the system's sensors (temperature sensor, servo motor position, water level sensor, water flow sensor)	24
6.2	Output of the circuit shows the vascular pressure and the heart rate of the patient on the OLED display	24

Dedicated to My Mom and My supervisor Mam

Chapter 1

Introduction

1.1 Background of project

Hemodialysis is a medical procedure used to filter waste and extra fluids from the blood when the kidneys can no longer perform this function adequately. Near 10% of the population worldwide is affected by chronic kidney disease (CKD), and millions die each year because they do not have access to affordable treatment [1]. This project aims to give these people a low cost and easily accessible dialysis treatment that is much needed for them. A portable hemodialysis machine can be accessible to anywhere, that the patient need not to move to the hospitals that are far away from home. The hemodialysis process is also very expensive for the victims. These project also provide a very low cost dialysis treatment for the patient compare to the existing hemodialysis device.

1.2 Objective of the project

The primary objective of portable hemodialysis project is to design and develop an innovative, user-friendly system that revolutionizes dialysis treatment for kidney disorder patients. The focus of this project lies in creating a compact and efficient solution that enhances patient mobility, thereby improving their quality of life. This portable system aims to offer the efficacy of traditional dialysis while ensuring ease of use, safety, and adherence to medical standards. By prioritizing a patient-centric approach and incorporating technological advancements, we aim to deliver a cost-effective solution that promotes patient compliance and acceptance.

Furthermore, recognizing its multifaceted utility, this project envisions this portable hemodialysis system as a versatile tool. Beyond its primary clinical application, it aim to design it to function as a trainer kit for educational labs. This dual-purpose functionality allows it to serve as an educational resource for healthcare professionals-in-training, enabling them to learn and practice essential techniques in a controlled environment. Through clinical validation, regulatory compliance, and comprehensive training programs, we strive to introduce a reliable and impactful portable hemodialysis system that addresses the unmet needs of patients and healthcare providers, potentially transforming the landscape of dialysis care

Chapter 2

Motivation/ Background Study

2.1 Urinary system

The **urinary system**, also known as the **urinary tract** or **renal system**, consists of the kidneys, ureters, bladder, and the urethra. The purpose of the urinary system is to eliminate waste from the body, regulate blood volume and blood pressure, control levels of electrolytes and metabolites, and regulate blood pH. The urinary tract is the body's drainage system for the eventual removal of urine. The kidneys have an extensive blood supply via the renal arteries which leave the kidneys via the renal vein. Each kidney consists of functional units called nephrons. Following filtration of blood and further processing, wastes (in the form of urine) exit the kidney via the ureters, tubes made of smooth muscle fibres that propel urine towards the urinary bladder, where it is stored and subsequently expelled from the body by urination (voiding). The female and male urinary system are very similar, differing only in the length of the urethra.

Urine is formed in the kidneys through a filtration of blood. The urine is then passed through the ureters to the bladder, where it is stored. During urination, the urine is passed from the bladder through the urethra to the outside of the body.

800–2,000 millilitres (mL) of urine are normally produced every day in a healthy human. This amount varies according to fluid intake and kidney function.

A simplistic mechanical analogy of the heart is that of a double-pump. The left and right sides are separate, but pump in unison to move blood through the heart [2].

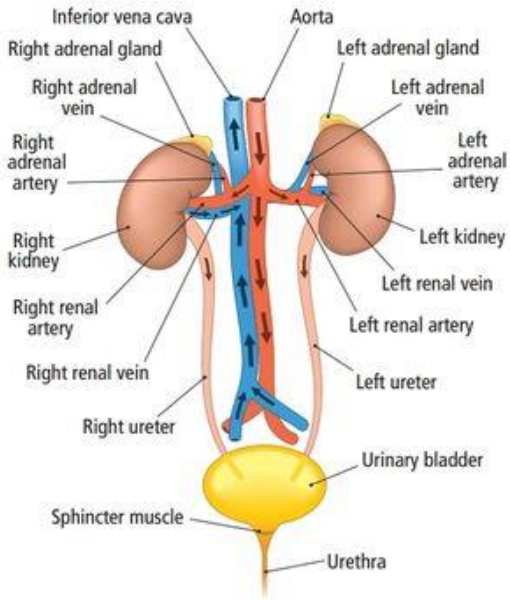


Fig. 2.1. Urinary System[3].

2.2 Mechanism of Wastages (Urine) Formation

Waste is often expelled by the human body in the form of urine, and urine is made in the nephron, the kidney's basic structural unit. Nephrons are important as they are the functional and structural units in the kidney. Nephrons, which filter the whole volume of blood in the body 60 times a day, are the backbone of the urinary system.

Glomerular filtration, **reabsorption**, and **secretion** are the three primary stages in the production of urine in our bodies .

2.2.1 Glomerular filtration

- The glomerulus is where blood is filtered. The Bowman's capsule epithelium, the endothelium of glomerular blood vessels, and the membrane between these two layers - all participate in this form of filtration.
- Glomerular filtration begins when the fluid fraction of blood is filtered by the glomerulus and enters the glomerular capsule as glomerular filtrate. The filtrate is a liquid that enters the nephron after leaving the bloodstream. Except for the formed elements and plasma proteins, it encompasses all chemicals in the blood

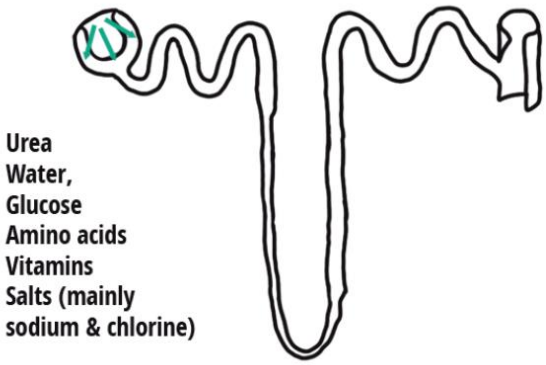


Figure 2.2: Filtration in Nephron

- Water, glucose, ions, amino acids, and nitrogenous wastes are all found in this filtrate, which is collected by the renal corpuscle.
- Water and other solutes are forced out of the circulation into Bowman's capsule by high blood pressure in the glomerulus. This high blood pressure is due to the afferent arteriole being more prominent than the efferent arteriole.
- It is a non-energy using method (ATP).
- Other capillary membranes are 1,000 times more permeable to water and solutes than the filtering membrane

2.2.2 Reabsorption

The reabsorption process takes place mostly in the renal tubules (the proximal convoluted tubule (PCT), a loop of Henle, and the distal convoluted tubule (DCT). This is where 99%

of the filtrate is produced. It refers to the return of substances from the filtrate (in the nephron) to the blood circulation (of peritubular capillaries). Urea, uric acid, and creatinine are nitrogenous wastes that are not reabsorbed. This is an active as well as a passive process.

- **The Proximal Convoluted Tubule (PCT):** The proximal convoluted tubules are where the majority of reabsorption takes place. The glucose in the filtrate is reabsorbed into the blood, most of which occurs in the PCT. The Na^+-K^+ pump, which employs ATP, actively transports most of the sodium cations (65%) from the proximal convoluted tubules back into circulation. Amino acids and vitamins are reabsorbed into the bloodstream from the PCT.

- **Loop of Henle:** The remaining water is reabsorbed by the descending limb. From the ascending limb, sodium ions and chloride ions are reabsorbed.

- **The Distal Convoluted Tubule (DCT):** DCT is capable of selective reabsorption of chemicals that have remained in the filtrate. It is dependent on the body's present needs. Hormones such as aldosterone regulate it. Also, aldosterone regulates sodium ions' reabsorption in DCT.

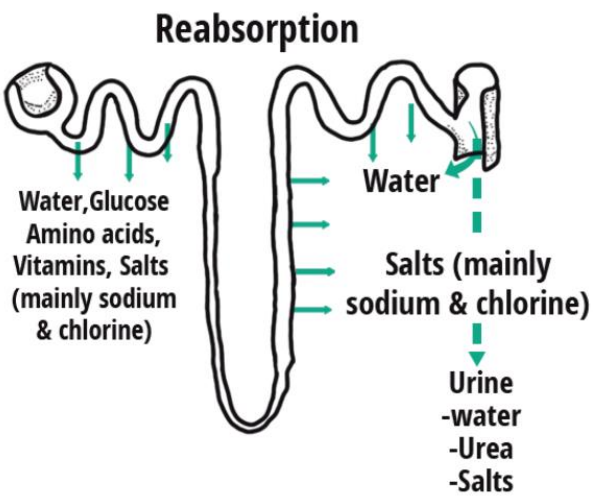


Figure 2.3: Tubular Reabsorption in the Nephron

2.2.3 Secretion

- It includes tubular secretion. It entails adding substances to the filtrate obtained from the blood in peritubular capillaries (by active transport).

- Any remaining nitrogenous wastes, some medications, and H^+ ions are the main materials secreted to maintain the pH of the blood.

- Tubular cells secrete hydrogen ions, potassium ions, and other ions into the filtrate.

- As Na^+ is reabsorbed, the potassium (K^+) cation is released into DCT. The hormone aldosterone regulates K^+ secretion.

- The pH of the blood rises as H^+ ions are released into the filtrate (it becomes less acidic).

- The secretion also removes urea, uric acid, ammonium ions, and certain medicines from the blood.
- Urine is the result of the fluid entering the collecting duct [4].

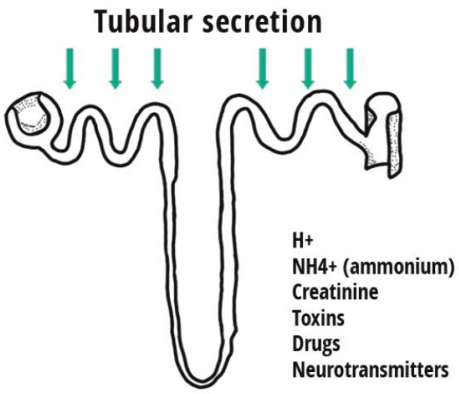


Figure 2.4: Tubular Secretion in Nephron

2.3 Dialysis

Dialysis is a way of cleaning body blood when kidneys can no longer do the job. It gets rid of body wastes, extra salt and water, and helps to control blood pressure. It is a life-saving technique for patients with End Stage Kidney Disease (ESKD) or Acute Kidney Injury.

People who have kidney failure, or end-stage renal disease (ESRD), may need dialysis. Injuries and conditions like high blood pressure, diabetes and lupus can damage kidneys, leading to kidney disease

Dialysis helps the body by performing the following functions of failed kidneys:

- Purification of blood by removal of waste products such as creatinine, urea etc.
- Removal of excess fluid and maintenance of the right amount of water in the body.
- Correction of electrolyte and acid-base balance disturbances.

2.3.1 History of Dialysis

The history of dialysis dates back to the 1940s. The first type of dialyzer, then called the artificial kidney, was built in 1943 by Dutch physician Willem Kolff. Kolff had first gotten the idea of developing a machine to clean the blood after watching a patient suffer from kidney failure. When his invention was completed, he attempted to treat over a dozen patients with acute kidney failure over the next two years. Although only one treatment turned out successful, he continued to experiment in improving his design.

Kolff came to the United States in the late 1940s and went to work at Mount Sinai hospital, trying to get kidney treatment to become a health service. Because of the hospital's opposition to the type of therapy at the time, he and his colleagues conducted treatment issues at a separate site. More experimentation lead to the manufacturing of an improved

design in the early 1950s. However, Kolff's device only treated acute kidney failure and not end stage renal disease (ESRD).

Dr. Belding Scribner, a professor of medicine at the University of Washington, developed a way for ESRD patients to received treatment through an access point in their arm. In 1962, Scribner opened the first official dialysis clinic for patients. He eventually developed a portable dialysis machine that allowed patients to receive dialysis treatment at home. By 1973, 40 percent of dialysis patients were doing their treatments at home.[5]

A German doctor by the name of Georg Haas, from the town of Giessen near Frankfurt am Main, performed the first dialysis treatments involving humans. It is believed that Haas dialyzed the first patient with kidney failure at the University of Giessen in the summer of 1924, after performing preparatory experiments.[6]

Figure 2.5: Dr. Georg Haas performing dialysis on a patient at the University of Giessen[7]

2.4 Types of Dialysis

2.4.1 Hemodialysis

In hemodialysis (HD), waste products and excess fluids are removed from the blood by passing the blood through a special filter or artificial kidney called a dialyzer, aided by a dialysis machine. (or, the process that involves the removal of chemical substances from the blood by passing it through tubes surrounded by semipermeable membrane).

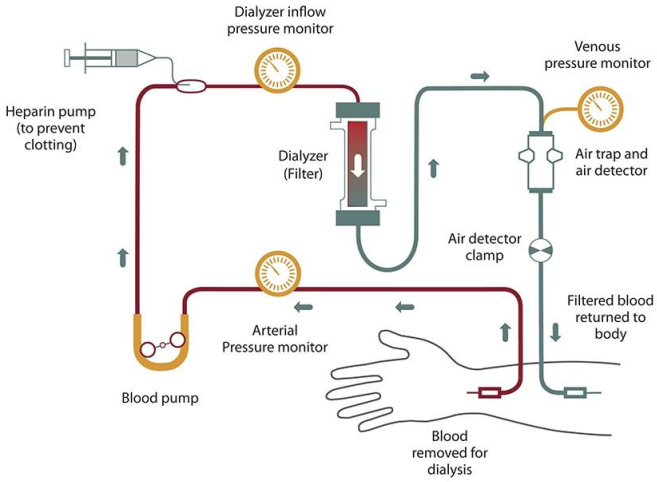


Figure 2.6: Hemodialysis Process [8]

2.4.2 Peritoneal dialysis:

Peritoneal dialysis Uses the peritoneum in a person's abdomen as the membrane through which fluid and dissolved substances are exchanged with the blood. It is used to remove excess fluid, correct electrolyte problems, and remove toxins in those with kidney failure.

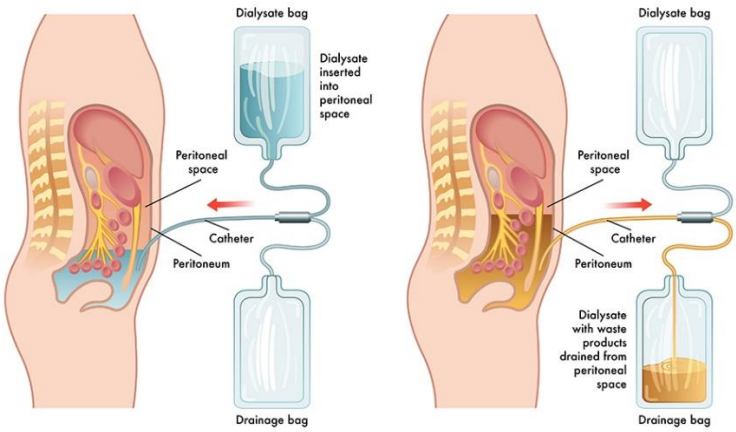


Figure 2.7: Peritonial Dialysis Process [9]

.

Chapter 3

Methodology

A Hemodialysis Machine consists of dialyzer, heater, some sensors (flow sensor, temperature sensor, bubble detector), blood pumps, hemodialysis blood tubing set.

3.1 Apparatus Required

Table 3.1 Components required for the System

SL No	Component Name	Description	Quantity	Cost(BDT)
1	Arduino mega	Microcontroller(2560 Rev3)	1	1450
1	Arduino Uno	Microcontroller(ATmega328P)	1	690
2	Power Supply	12V-2A,9V-2A,5V-2A	3	330
3	OLED Display	SSD1306	1	350
3	Dialyzer	ELISIO™-H	1	1200
4	Hemodialysis Blood Tubing Set		1	260
5	Heater		1	80
6	Temperature Sensor	DS18B20	2	600
7	Water Flow Sensor	YF-S201	1	380
8	Water level Sensor	DC3-5V	1	70
9	Pulse Sensor	-	1	150
10	Air Pressure Sensor	HX710B	1	800
11	Pump	12V-2A	2	700
13	Relay	1 Channel 5V relay	1	90
14	Servo Motor	MG996R	1	350
16	Jumper Wire	M-M,M-F,F-F	As required	270
17	Fabricated 3D Heparin pump Casing	-	1	300
17	Injection	5ml	1	5
18	Pipe	2 meter	1	40
	Total			8115

3.2 Architecture of the Hemodialysis Machine

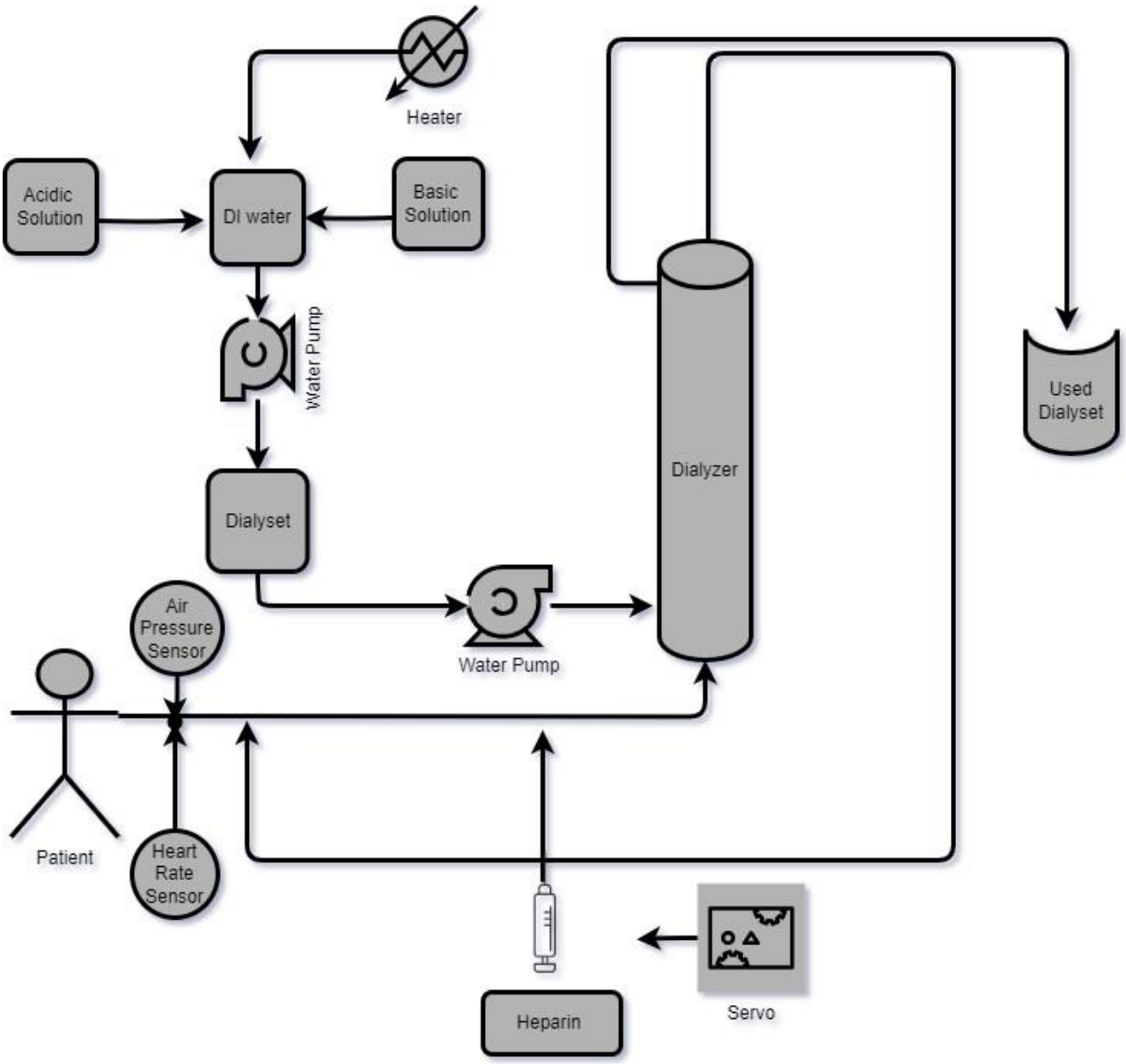


Figure 3.1: Functional block diagram of the Hemodialysis Machine.

3.3 Flowchart

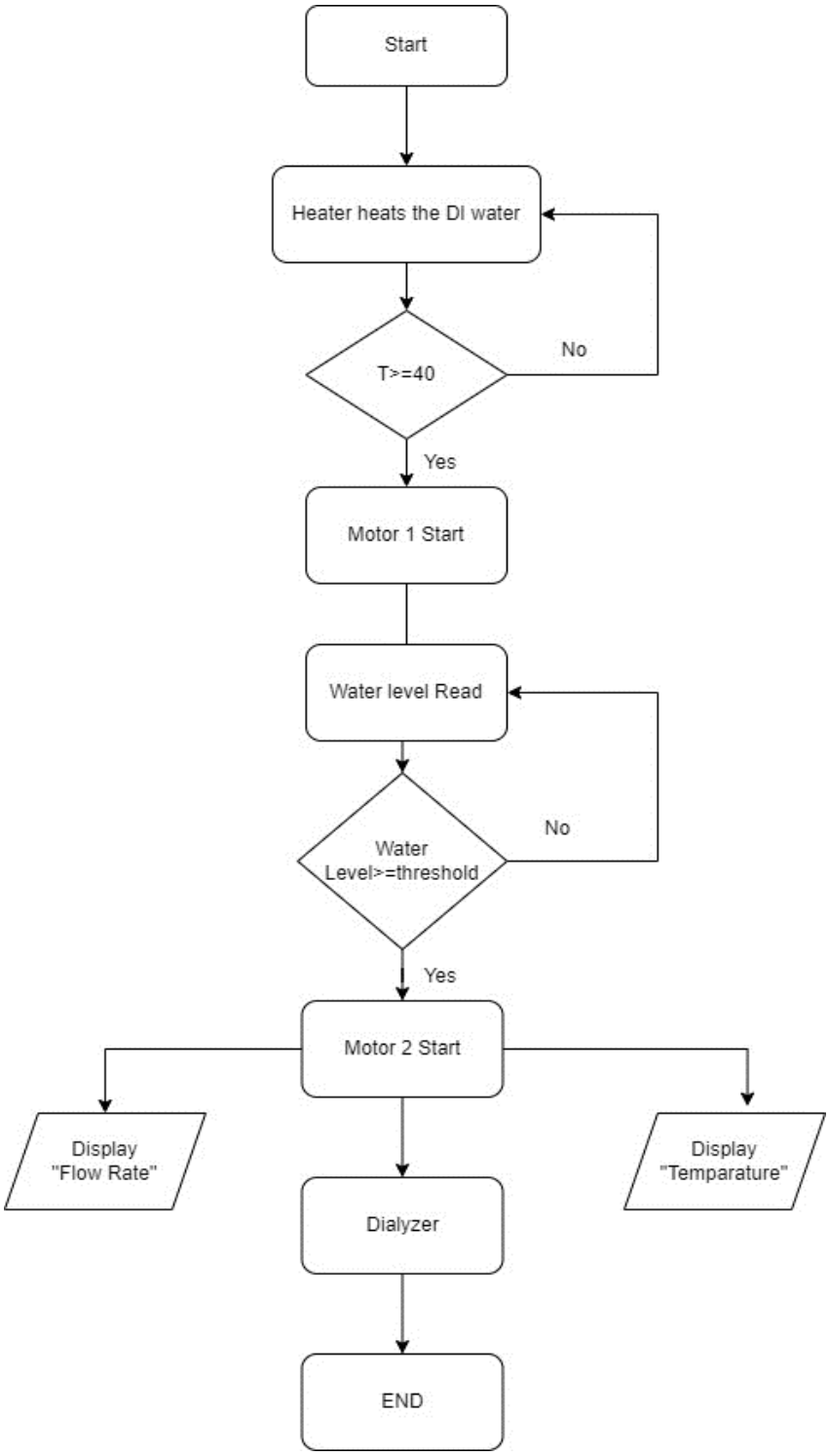


Figure 3.2: Desired Flowchart of the System

Chapter 4

Design procedure/Experimental set-up/ Circuit Diagram

4.1 Design procedure

1. Heater heats the DI water to a specific temperature for removing microorganism and the first motor on

2. The first motor passes the DI water with the acidic, basic and 5% dextrose solution to the

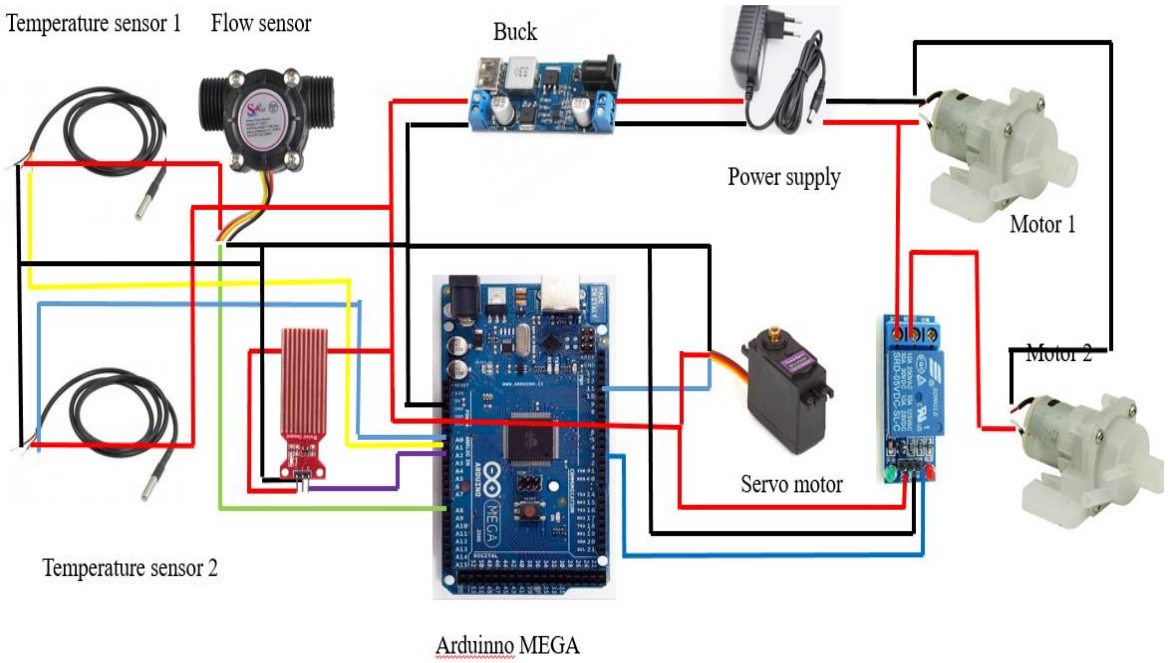
3. The dialysate solution mixed at a threshold level and the second motor starts pumping dialysate

4. Wastages are drained from the dialyzer to a waste jar.

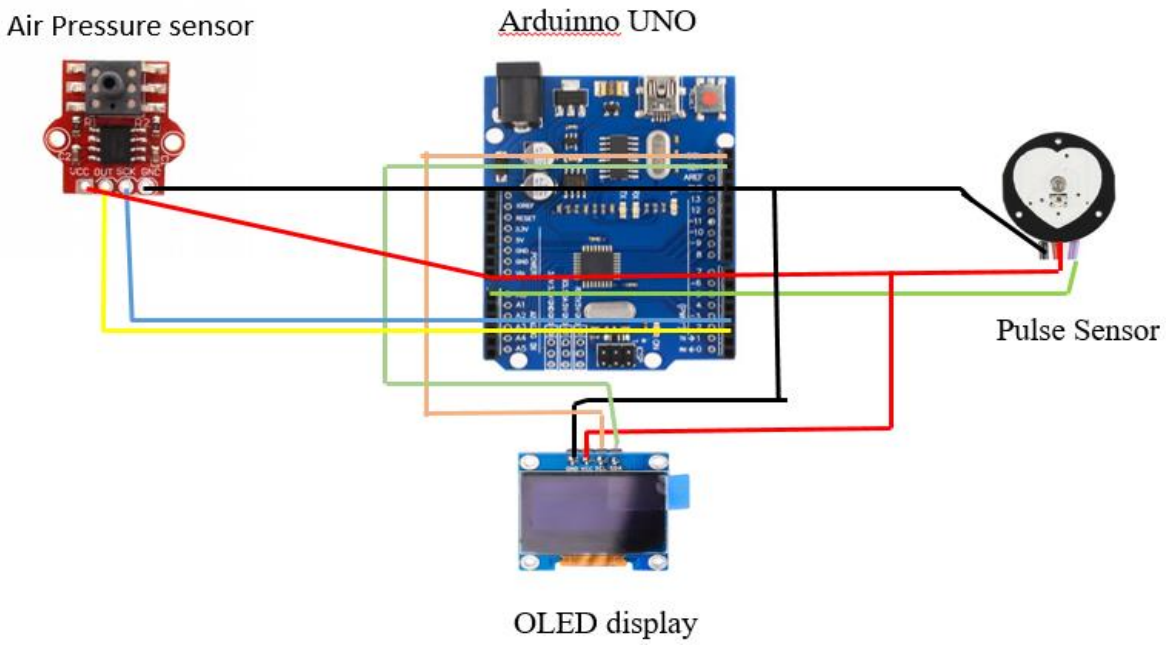
5. Arterial pressure, venous pressure and Heart rate were measured.

Figure 4.1: Design Procedure of the System

4.2 Circuit Diagram



(a)



(b)

Figure 4.2: Overall circuit diagram of the system,(a) for the dialysate mixing and pumping process (b) fir measuring the heart rate and the arterial, venous pressure.

Chapter 5

Implementation

5.1 Circuit Implementation (step by step)

The step by stem implementations of the system's full circuit are:

1. Two temperature sensors are connected to A0 and A1 pin of Arduino Mega.
2. Water level senor connected to the A2 pin that used for turning on the motor using relay.
3. Water flow sensor connected to A8 pin that to the dialyzer.
4. Servo connected to the pin 11 and also powered by a 5v external power source
5. The extended section used Arduino Uno where the Heart Rate Sensor connected to A0 and pressure sensor's OUT and SKT pin connected to 2 and 3 pin.

5.2 Circuit Implementation of the main system.

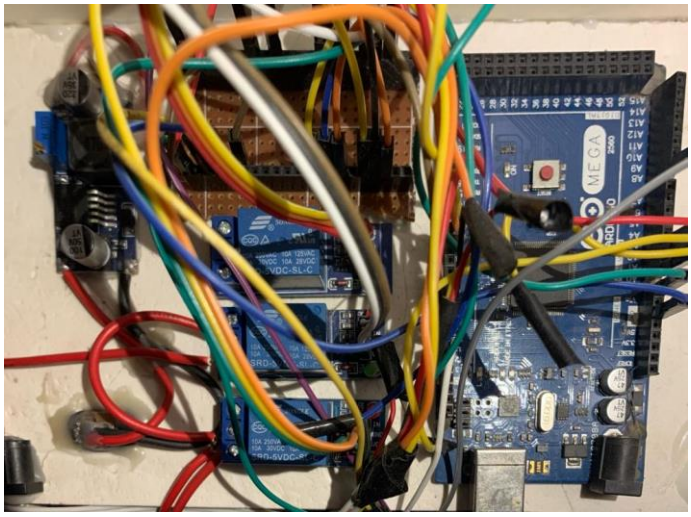


Figure 5.1: Circuit for the dialysate mixing and pumping process

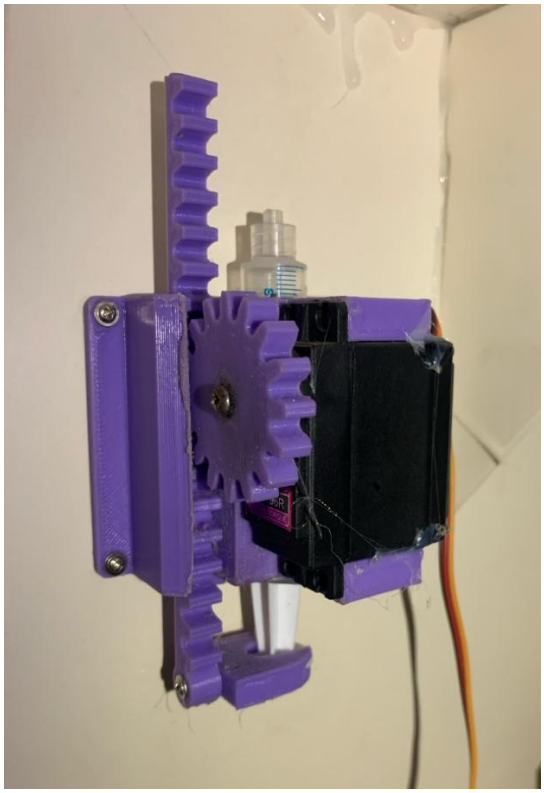


Figure 5.2: Implementation of the smart Heparin pump using servo motor.

Figure 5.3: Setup of the Dialyzer to the system.

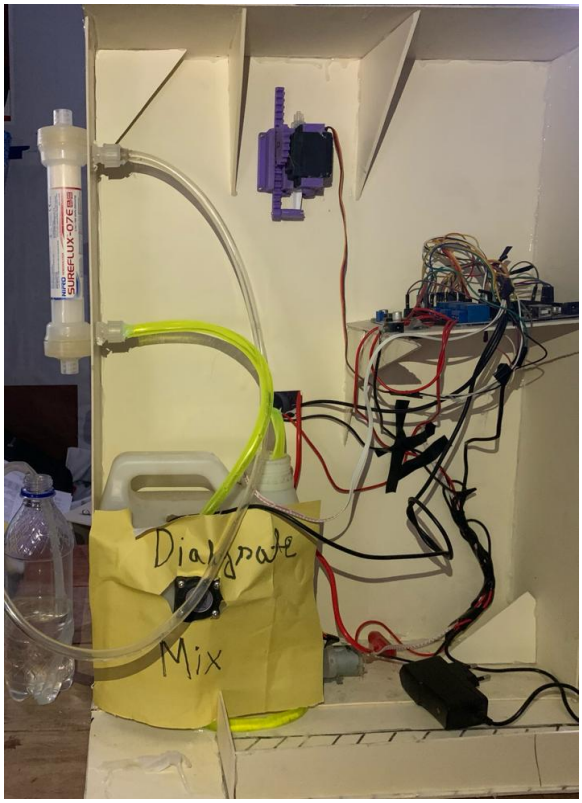


Figure 5.4: Final implementation of the project (front view).

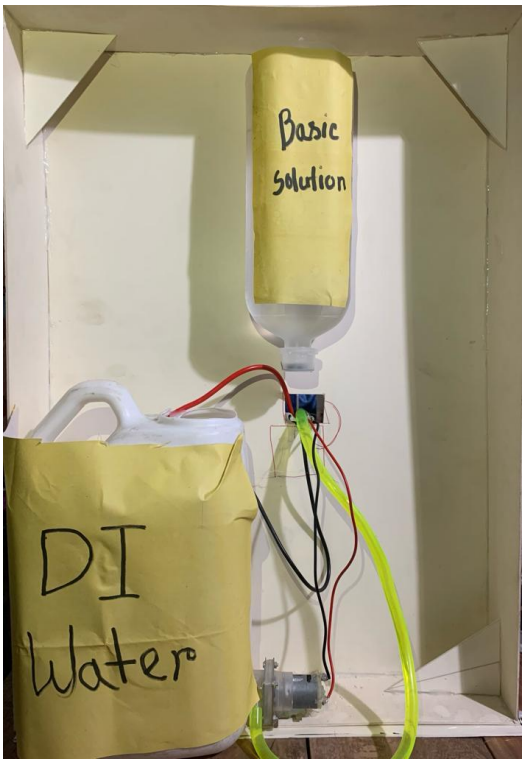


Figure 5.5: Final implementation of the system (back view).

5.2 Implementation of the extended part

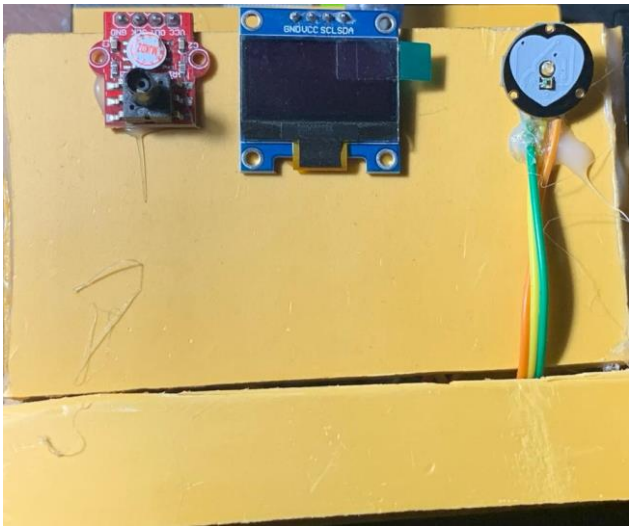


Figure 5.6: Extended part that measure the heart rate and vascular pressure.

Chapter 6

Results and Discussions

6.1 Results

6.1.1 Main Circuit

The output of the sensors from the main circuit are shown in the serial monitor of the Arduino IDE software given in the figure (6.1).

```
Temperature from sensor 1: 25.50°C
Temperature from sensor 2: 26.19°C
Water Level: 64
Flow rate: 660.00 mL/min
Servo Position: 164
Temperature from sensor 1: 25.50°C
Temperature from sensor 2: 26.19°C
Water Level: 65
Flow rate: 659.00 mL/min
Servo Position: 162
```

Figure 6.1: Output of the system's sensors (temperature sensor, servo motor position, water level sensor, water flow sensor)

6.1.2 Extended circuit

The extended part of the circuit shows the vascular pressure and the heart rate of the patient in figure (6.2)

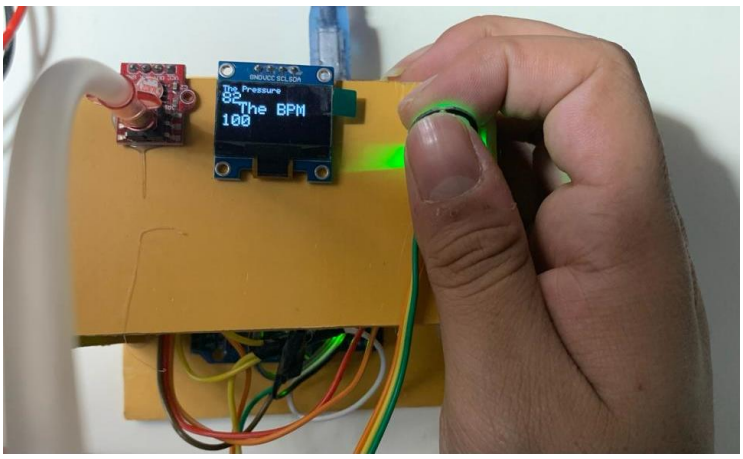


Figure 6.2: Output of the circuit shows the vascular pressure and the heart rate of the patient on the OLED display

6.2 Discussions

The result obtained from the serial monitor of the Arduino IDE software gives the output of the sensors that are used in the main circuit aka the dialysate mixing and pumping part in figure (6.1). It gives the readings of the two sensors that are placed in the DI water jar and the dialysate mixing jar. The first temperature sensor (DS18B20) mainly shows the temperature of the DI water that the heater heats. The second one used for measuring the temperature of the dialysate mixture jar that will pass from the dialyzer. The water level sensor measures the level of water which will define that the dialysate mixture completes or not. If the level of the water cross the threshold then the relay turned on the second motor. The water flow sensor measure the flow rate of dialysate that passes to the dialyzer. The output also shows the servo motor level that automatically pumps heparin to the blood tubing set.

The extended part of the system used two sensor shows the vascular pressure and the heart rate of the sensor in figure (6.2).In case of measuring the vascular pressure, the blood tube part of the blood tubing set attached to the air pressure sensor. For measuring the heart rate, the finger tip positioned on the heart rate sensor.

Chapter 7

Conclusion and Future Work

7.1 Conclusion

The culmination of our efforts in developing the portable hemodialysis system represents a pioneering stride in modern healthcare. Our commitment to innovation, patient-centric design, and technological advancement has yielded a groundbreaking solution. This compact, efficient system not only enhances the quality of life for kidney disorder patients through improved mobility and treatment efficacy but also stands as a testament to our dedication to accessible and affordable healthcare. The dual functionality of serving as a clinical tool and a training resource amplifies its impact, underscoring its versatility in transforming both patient care and medical education. As we move forward, our focus remains resolute: to continually push boundaries, bridge gaps in healthcare access, and set new standards in dialysis care. This project embodies our ethos of innovation and compassion, striving not only for technological excellence but also for meaningful improvements in the lives of those we aim to serve.

7.2 Future Work

In future iterations, the focus will extend towards enhancing the portability and efficiency of the system. Integrating battery-powered functionality will render the system completely portable, allowing for unhindered use regardless of location. Moreover, the goal is to further compact the design while elevating its efficiency, ensuring a streamlined and potent solution for dialysis care. Continuous efforts will be directed towards optimizing the system's performance, aiming for higher efficacy and reliability, thus solidifying its position as a game-changer in portable dialysis technology.

In addition, the vision includes repurposing the system as a demonstration kit for educational institutes. By leveraging its multifunctional design, this innovation will serve as an invaluable tool in educational labs, offering hands-on training opportunities for aspiring healthcare professionals. This dual utility, as both a portable dialysis system and an educational demo kit, underscores its versatility and potential impact, fostering learning and skill development in the next generation of medical practitioners.

References

[1] *Global facts: About kidney disease* (2023) *National Kidney Foundation*. Available at: https://www.kidney.org/kidneydisease/global-facts-about-kidney-disease (Accessed: 01 December 2023).

[2] Park, J.B. (2003) *Biomaterials: Principles and applications*. Boca Raton, 125 London Wall, London EC2Y 5AS, United Kingdom: CRC Press.

[3] Wiki, C. to C. (no date) *Urinary system, Classification Wiki*. Available at: https://classification.fandom.com/wiki/Urinary_system?file=9737073.jpg (Accessed: 02 December 2023).

[4] Vedantu (2023) *Mechanism of urine formation - important concepts for NEET, VEDANTU*. Available at: https://www.vedantu.com/neet/mechanism-of-urine-formation (Accessed: 01 December 2023).

[5] Bracamonte, H. (2020) *A brief history of dialysis, Dialysis Patient Citizens Education Center*. Available at: https://www.dpcedcenter.org/news-events/news/a-brief-history-of-dialysis/ (Accessed: 01 December 2023).

[6] *The history of dialysis Fresenius Medical Care*. Available at: https://www.freseniusmedicalcare.com/en/media/insights/company-features/the-history-of-dialysis (Accessed: 02 December 2023).

[7] *Fresenius Medical Care*. Available at: https://www.freseniusmedicalcare.com/fileadmin/data/com/pdf/Media_Center/Publications/Annual_Reports/FME_Annual_Report_2019.pdf (Accessed: 01 December 2023).

[8] *An overview guide for dietitians - national institute of diabetes and ...* Available at: https://www.niddk.nih.gov/-/media/Files/Health-Information/Health-Professionals/Kidney-Disease/ckd-diet-assess-manage-treat-508.pdf (Accessed: 01 December 2023).

[9] *Peritoneal dialysis* (2023) *HealthCommunities Provider Services*. Available at: https://www.healthcommunitiesproviderservices.com/patient-education/peritoneal-dialysis/ (Accessed: 02 December 2023).

Appendix A

Arduino IDE code for the main circuit of the system-

```
#include <OneWire.h>
#include <DallasTemperature.h>
#include <Servo.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>
#define SCREEN_WIDTH 128 // OLED display width,  in pixels
#define SCREEN_HEIGHT 64 // OLED display height, in pixels
int servoPosition = 180;
#define BUTTON_PIN 12
#define RELAY1 2
#define RELAY2 3// MOTOR1_RELAY_PIN
#define RELAY3 4 //motor1
#define MOTOR2_RELAY_PIN 10 //
int pos;

#define ONE_WIRE_BUS_1 A0 // Temperature sensor 1
#define ONE_WIRE_BUS_2 A1 // Temperature sensor 2
#define WATER_LEVEL_SENSOR_PIN A2 // Pin for the water level sensor (analog
input)
const int flowPin = A8;

volatile int flowCounter; // Counter for the number of pulses from the flow sensor
float flowRate;          // Calculated flow rate in L/min
float Temp1;
float Temp2;
int waterLevel; // Water level sensor reading
int buttonState=0;
// Setup two separate OneWire instances
OneWire oneWire1(ONE_WIRE_BUS_1);
OneWire oneWire2(ONE_WIRE_BUS_2);

// Pass each OneWire reference to separate DallasTemperature instances
DallasTemperature sensor1(&oneWire1);
DallasTemperature sensor2(&oneWire2);

// Servo motor control
Servo myservo;

// Interrupt Service Routine (ISR) to count pulses
```

```
void flowPulseCounter()
{
  flowCounter++;
}

void setup()
{

  pinMode(BUTTON_PIN, INPUT);
  // Start serial communication
  Serial.begin(9600);
  pinMode(RELAY1, OUTPUT);
  pinMode(RELAY2, OUTPUT);
  pinMode(RELAY3, OUTPUT);
  pinMode(MOTOR2_RELAY_PIN,OUTPUT);
  //pinMode(flowpin,OUTPUT);

      //digitalWrite(MOTOR2_RELAY_PIN, HIGH);

  myservo.attach(11); // servo d11 a set korbi

  // Start up each sensor
  sensor1.begin();
  sensor2.begin();

  // Configure the flowPin as INPUT_PULLUP
  pinMode(flowPin, INPUT_PULLUP);

  // Attach the ISR to the flowPin and set it to RISING
  attachInterrupt(digitalPinToInterrupt(flowPin), flowPulseCounter, RISING);
  digitalWrite(RELAY2, HIGH);
}

void loop()
{
servoControl();
   //Serial.println(pos);

      // Serial.println(buttonState);
  // Request temperatures from each sensor
  sensor1.requestTemperatures();
  sensor2.requestTemperatures();
```

```
  // Read water level sensor
  waterLevel = analogRead(WATER_LEVEL_SENSOR_PIN);
 buttonState = digitalRead(BUTTON_PIN);
  // Print temperature from the first sensor
  Serial.print("Temperature from sensor 1: ");
  Serial.print(sensor1.getTempCByIndex(0));
  Temp1 = sensor1.getTempCByIndex(0);
  Serial.println("°C");

  // Print temperature from the second sensor
  Serial.print("Temperature from sensor 2: ");
  Serial.print(sensor2.getTempCByIndex(0));
  Serial.println("°C");
  Temp2 = sensor2.getTempCByIndex(0);

  // Print water level information
  Serial.print("Water Level: ");
  Serial.println(waterLevel);

  // Delay for a second
  delay(300);

  // Detach the interrupt to safely manipulate the flowCounter
  detachInterrupt(digitalPinToInterrupt(flowPin));

  // Calculate the flow rate
  flowRate = (flowCounter / 1.5); // (Pulse frequency x 60 seconds) / 7.5 Q, where Q is the
factor of the flow sensor (7.5 for the YF-S201 sensor)

  Serial.print("Flow rate: ");
  Serial.print(flowRate); // Print the flow rate to the serial monitor
  Serial.println(" L/min");

  flowCounter = 0; // Reset the flow counter

  // Reattach the interrupt
  attachInterrupt(digitalPinToInterrupt(flowPin), flowPulseCounter, RISING);

  // Call the relay control function
  relay();
}

void relay()
```

```
{
  // if (Temp1 >= 30 || buttonState ==1)
  // {

  //   digitalWrite(RELAY1, LOW);
  //   digitalWrite(RELAY2, HIGH);
  // }
  // else
  // {
  //   digitalWrite(RELAY1, HIGH);
  // }

  if (waterLevel >= 300)
  {
    digitalWrite(MOTOR2_RELAY_PIN, HIGH);
    digitalWrite(RELAY2, LOW);


  }
  /*else
  {
    digitalWrite(MOTOR2_RELAY_PIN, LOW);


  }*/
}
void servoControl() {



  static unsigned long previousMillis = 0; // Holds the last time the servo moved
  const long interval = 10; // servo move delay
 // Initial servo position

  unsigned long currentMillis = millis();

  if (currentMillis - previousMillis >= interval) {
    previousMillis = currentMillis; // Save the last time we moved the servo

    servoPosition =  servoPosition-2; // Increment the servo position by 2 degrees

    if (servoPosition < 2) { // If the servo position is greater than 180 degrees, reset it
      servoPosition = 180;
```

```
    delay(100);
  }

  myservo.write(servoPosition); // Move the servo to the new position
  Serial.print("Servo Position: "); // Optional: Printing the current servo position
  Serial.println(servoPosition);
  pos=servoPosition;
  // Serial.println(pos);
  // Serial.println("move");

 }
}
```

Appendix B

Arduino IDE code for the extended circuit of the system-

```
// marge code
#include <Wire.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>
#define SCREEN_WIDTH 128 // OLED display width,  in pixels
#define SCREEN_HEIGHT 64 // OLED display height, in pixels
Adafruit_SSD1306 oled(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire, -1);
const int PIN_CLK = 3;
const int PIN_DATA = 2;
const int PIN_LED = 13; // This will also be used for the pulse sensor blink

// Function prototypes for Pressure Sensor
void initPressureSensorGPIO();
bool isPressureDataReady();
void sendPressurePulse();
```

```
unsigned long readPressureData();

// BPM Code
#define USE_ARDUINO_INTERRUPTS true
#include <PulseSensorPlayground.h>
#include <Wire.h>

// Variables for BPM
const int PulseWire = 0; // PulseSensor PURPLE WIRE connected to ANALOG PIN 0
int Threshold = 550;

PulseSensorPlayground pulseSensor; // Creates an instance of the PulseSensorPlayground
object called "pulseSensor"

void setup() {
  // Initialize Serial communication
  Serial.begin(9600);
   if (!oled.begin(SSD1306_SWITCHCAPVCC, 0x3C)) {
    Serial.println(F("SSD1306 allocation failed"));
    while (true);
  }
  // Initialize Pressure Sensor GPIO
  initPressureSensorGPIO();

  // Configure the PulseSensor object
  pulseSensor.analogInput(PulseWire);
  pulseSensor.blinkOnPulse(PIN_LED); // Use the LED for pulseSensor and pressure alert
  pulseSensor.setThreshold(Threshold);

  if (pulseSensor.begin()) {
    Serial.println("PulseSensor initialized.");
  }
   oled.clearDisplay();
}

void loop() {
  oled.clearDisplay();
  // Pressure Sensor routine
  unsigned long pressureData = readPressureData();
  pressureData = (pressureData / pow(2, 24)) * 900;
  Serial.print("Pressure Data Value: ");
  Serial.println(pressureData);
    oled.display();
```

```
  oled.setTextSize(1.3);          // text size
  oled.setTextColor(WHITE);     // text color
  oled.setCursor(0, 0);        // position to display
  oled.println("The Pressure ");
  oled.setTextSize(2);
  oled.println(pressureData);// text to display
  oled.display();

  if (pressureData > 10000) {
    // If the pressure is high, keep the LED on
    // Otherwise, let the PulseSensor control the LED
    digitalWrite(PIN_LED, HIGH);
  } else {
    // BPM routine
   // if (pulseSensor.sawStartOfBeat()) {
      int myBPM = pulseSensor.getBeatsPerMinute();
      myBPM = map(myBPM, 0, 270, 0, 115);
      Serial.println("♥ A HeartBeat Happened!");
      Serial.print("BPM: ");
      Serial.println(myBPM);
   oled.display();
  oled.setTextSize(2);          // text size
  oled.setTextColor(WHITE);     // text color
  oled.setCursor(20, 20);        // position to display
  oled.println("The BPM ");
  oled.println(myBPM);// text to display
  oled.display();
    //}
  }
  delay(200); // Delay for stability
}

void initPressureSensorGPIO() {
  pinMode(PIN_LED, OUTPUT);
  pinMode(PIN_DATA, INPUT);
  pinMode(PIN_CLK, OUTPUT);
  digitalWrite(PIN_CLK, LOW);
}

bool isPressureDataReady() {
  return digitalRead(PIN_DATA) == LOW;
}
```

```
void sendPressurePulse() {
  digitalWrite(PIN_CLK, LOW);
  delayMicroseconds(1);
  digitalWrite(PIN_CLK, HIGH);
  delayMicroseconds(1);
  digitalWrite(PIN_CLK, LOW);
}

unsigned long readPressureData() {
  unsigned long value = 0;
  while (!isPressureDataReady()) {
    // Waiting for sensor to become ready
  }
  for (int i = 0; i < 24; i++) {
    sendPressurePulse();
    value = value << 1;
    if (digitalRead(PIN_DATA)) {
      value++;
    }
  }
  sendPressurePulse(); // Skip the 25th pulse
  return value;
}
```